Automatic GFL/GFM Mode Switching

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Problem, research question, and contributions

The operating problem

GFL converters need an external angle reference; GFM converters can create one, but reserving every converter for GFM duty is costly and can introduce adverse mode interaction. When the last synchronous reference trips, the operator must decide more than simply "GFL or GFM."

How many should form, which ones, and by what transition route?

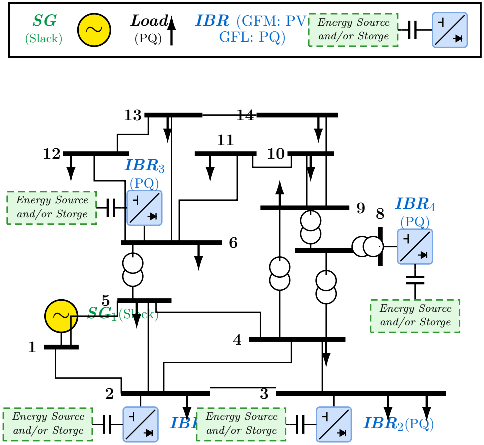
Two findings drive the design: linear admissibility alone does not guarantee that the present state can reach a configuration, and benefit is not monotone in the number of GFM units.

What this project delivers

1. One unified mathematical path from AC power flow through SSSA to switched time-domain simulation.
2. A dual-mode converter that follows or forms the grid, switching through a current-continuity-gated assignment.
3. Exhaustive screening of all $2^4 - 1 = 15$ non-empty GFM sets at their own equilibria.
4. A six-arm study on one 250 s chronology: zero forming units are refused, four pinned units collapse, one or two survive *without* adaptation, and staged switching holds the smallest *settled* frequency deviation on every window after the first — without pre-committing a selection.

Motivation: Cui et al., *IEEE TPS*, 2025; Ding et al., *IEEE TSTE*, 2025.

Study network and the dual-mode converter



The machine at bus 1 holds the healthy-system angle reference; buses 2, 3, 6, 8 carry four dual-mode converters. After the trip a certified grid-forming converter must own the island reference.

One coordinate layout, two regimes — exclusive

grid-following	10 active	7 controller states frozen
grid-forming	11 active	6 controller states frozen
tripped	0 active	everything held

17 coordinates per converter; the composite system carries $6 + 4 \times 17 = 74$. The 17-vector is a *superset*, not a simultaneously active state: only the selected controller branch evolves in each regime, and the other is held.

A mode change is an assignment

It happens at a frozen instant, not across a time step: the arriving branch is re-initialised at the present terminal operating point. The parameterisation may jump; the current the network sees may not.

The framework in four equations

Network equilibrium and bus semantics

$$F_{\mathrm{PF}}(z) = 0, \qquad J_{\mathrm{PF}}\Delta z = -F_{\mathrm{PF}},$$
$$g(x, V) = Y_{\mathrm{net}}V - I_{\mathrm{bus}}(x, V) = 0.$$

REF fixes $|V|$, θ; PV fixes P, $|V|$; PQ fixes P, Q. The same algebraic row set closes the network at every operating point below.

Exact DAE linearisation

$$\begin{bmatrix}\Delta\dot{x}\\ 0\end{bmatrix} = \begin{bmatrix}f_x & f_y\\ g_x & g_y\end{bmatrix}\begin{bmatrix}\Delta x\\ \Delta y\end{bmatrix}, \qquad A = f_x - f_y(g_y \backslash g_x).$$

One network-angle gauge coordinate is removed. Paired left/right eigenvectors give participation; $\zeta = -\sigma/\sqrt{\sigma^2 + \omega^2}$ is compared with the declared 0.05 floor.

One coupled implicit time step

$$R_x = x_{n+1} - x_n - \frac{h}{2}[f(x_n, y_n) + f(x_{n+1}, y_{n+1})] = 0,$$
$$R_y = g(x_{n+1}, y_{n+1}) = 0.$$

Damped Newton solves both residuals together, and accepted steps land exactly on the scheduled events.

A switch is an assignment, not a step

$$\left\|I^+ - I^-\right\|_\infty \leq 10^{-10}\ \text{p.u.}$$

The commit also requires the right-limit network solve to converge at the new configuration; otherwise nothing is published.

Mathematical bases: IEEE Std 1110; Kundur (1994); Kodsi–Cañizares (2003); MATPOWER IEEE-14 data; stated project derivations.

When does it switch?

Normalised severity

$$J_V = \frac{\big||V_i| - V_i^\star\big|}{0.10\,\mathrm{pu}}, \qquad J_f = \frac{|f_{\mathrm{COI}} - 60|}{0.50\,\mathrm{Hz}},$$

$$S = \mathrm{sat}_{[0,1]}\left(\tfrac{1}{2}J_V + \tfrac{1}{2}J_f\right).$$

$V_i^\star$ is the healthy power-flow voltage, so J_V answers to a local collapse or overshoot and J_f to a system-wide excursion. The equal weighting was frozen before any result was read.

Asymmetric trigger

$$S \geq 0.65 \text{ held } 0.10\,\mathrm{s} \qquad (\mathrm{GFL} \rightarrow \mathrm{GFM}),$$
$$S < 0.35 \text{ held } 1.00\,\mathrm{s} \qquad (\mathrm{GFM} \rightarrow \mathrm{GFL}).$$

The gap prevents chatter; the unequal dwell makes support fast and release deliberately slow.

What must hold before a switch is accepted

The candidate configuration must already be certified at its own equilibrium; the arriving state is then carried forward with the same simulator over the destination’s own settling horizon. A trial that would lose synchronism is refused, the present mode is kept, and unresolved transitions fail closed. No parameter, event or acceptance threshold is ever changed in response to the observed curves.

One validation, three analyses GATED

Steady state vs PSAT 2.1.11, identical mapped inputs

quantity	IEEE 14	RTS-24
network contract max $\lvert\Delta Y\rvert$	3.97×10^{-15}	2.84×10^{-14}
power flow max $\lvert\Delta V\rvert$ [pu]	6.661338×10^{-16}	4.441×10^{-16}
power flow max $\lvert\Delta\theta\rvert$ [deg]	3.552714×10^{-14}	6.750×10^{-14}

Tolerances declared *before* the comparison: $\Delta V < 10^{-6}$ pu, $\Delta\theta < 10^{-4}$ deg. The gate passes with ten orders of magnitude to spare — the two solvers agree at machine precision, not merely within a tolerance.

Transient trajectories vs PSAT 2.1.11, IEEE 14-bus

metric	max difference	gate
$\Delta\delta_{\mathrm{COI}}$ [deg]	9.571×10^{-3}	< 0.05
$\Delta\omega$ [pu]	3.816×10^{-6}	$< 1.0 \times 10^{-4}$
ΔP_e [MW]	4.217×10^{-2}	< 0.10
$\Delta\lvert V\rvert$ (fault bus) [pu]	3.179×10^{-5}	$< 1.0 \times 10^{-3}$

Every pre-declared gate is satisfied over the full 15 s horizon, with zero unconverged steps.

Small signal, five-machine classical IEEE 14. All four oscillatory modes agree to $\lvert\Delta\lambda\rvert = 1.000 \times 10^{-6}\ \mathrm{s}^{-1}$ — a uniform, documented regularisation offset of the reference analysis — and the physical frequencies match to the printed precision (1.83350, 1.58440, 1.53137, 1.35374 Hz on both sides). The same gate *refuses* a second third-party tool, and that refusal is printed rather than hidden. This is the independent baseline established on classical machines *before* the converters enter.

The spectrum, at the two operating points that matter

Machine online, all converters following

44 physical coordinates of 45 (one angle removed by the network-angle quotient). The electromechanical mode is the one the case contract speaks about:

λ	$-1.146045 \pm 7.254946j$
f	1.154661 Hz
ζ	0.156033
dominant state	machine speed

against the declared floor $\zeta_{\min} = 0.05$ — a factor of three of margin, and the smallest damping ratio in the table.

Machine tripped, one converter forming

40 physical of 41; 30 printed rows, conjugate pairs once; *entirely* in the left half plane — the island the converters hold is small-signal stable.

	λ	dominant
fastest	-2075.371442	inner current
DC family	$-98.41 \pm 98.38j$	DC link
	. . . 3 more pairs	DC link
PLL pair	$-0.588912 \pm 2.171749j$	PLL angle
slowest	$-0.328014 \pm 0.448860j$	voltage loop

The slowest root *is* the decay-rate margin the selector ranks on.

Read with two limits, stated rather than buried OPEN

With the machine online the slowest root is $-0.065196 \pm 0.124395j$ (0.019798 Hz, $\zeta = 0.464215$, machine speed) and the slowest real decay rate is -0.174920 (machine field flux). In the island table, 9 of the 30 rows have their two leading participations within 0.02, so the named state there is *not* a sole attribution. And each table is one linearisation point: it is not, and must not be read as, a stability certificate for the switched trajectory.

Predicted first, then measured: the DC link

What was predicted, before the run

The converter's DC side was an ideal source. Replacing it by a real source behind a resistance and an inductance adds one state per converter, and its time constant is not free: requiring the flattest possible response fixes the damping at $\zeta^\star = 1/\sqrt{2}$ and therefore the time constant at 5.00 ms. From that alone the pole pair was predicted, on paper, at

$$-97.7 \pm 97.7\,\mathrm{s}^{-1}, \quad f \approx 15.5\,\mathrm{Hz}, \quad \zeta = 0.7071.$$

Two feasibility bounds were declared at the same time, and the realised margin against the binding one is 2.07–2.15. The source voltage is not chosen either — the dispatch forces it, giving 1.0269–1.0448 p.u. across the four converters.

What the assembled system then returned GATED

	predicted	measured
f [Hz]	≈ 15.5	15.18–15.66
ζ	0.7071	0.7072–0.7079

Four converters, four pairs, every one of them where the derivation said it would be. Participation splits exactly 0.500/0.500 between the two DC coordinates and is 0.000 everywhere else, so each pair *is* one converter's DC circuit and nothing else.

And the coupling was measured, not assumed OPEN

The AC equations read the DC states with sensitivity *exactly zero*; the DC equations do read the AC. The coupling is one-way — which also means a sagging DC link never limits an AC command in this model, so nothing here covers DC-limited ride-through.

How many converters, and which — computed

Candidate Set Admissibility

size	grid-forming set	margin [s^{-1}]	outcome
1	bus 8	−0.155437	admissible
1	bus 6	−0.252025	admissible
1	bus 3	−0.316711	admissible
1	bus 2	−0.328014	admissible
2	buses 2 + 6	−0.561723	admissible
2	**buses 3 + 6**	**−0.568138**	**best**
2	four other pairs	–	limit hit
3	all four triples	–	no equilibrium
4	all converters	−0.482909	admissible

What the enumeration shows

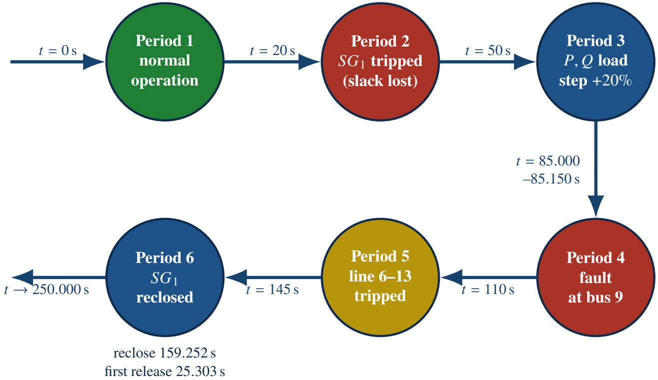
15 candidate sets evaluated, 7 admissible — 4 of 4 at size 1, 2 of 6 at size 2, 0 of 4 at size 3, 1 of 1 at size 4.

- **Non-monotonic**: single sets pass, every triple fails, and the best margin sits at a pair ($-0.56814\,\mathrm{s}^{-1}$).
- **Best pairing**: buses 3 + 6, ahead of any single unit and of the full four-converter set.
- **Computed, not assumed**: sizing cannot follow a rule of thumb — each set is screened at its own equilibrium.

Necessary, not sufficient OPEN

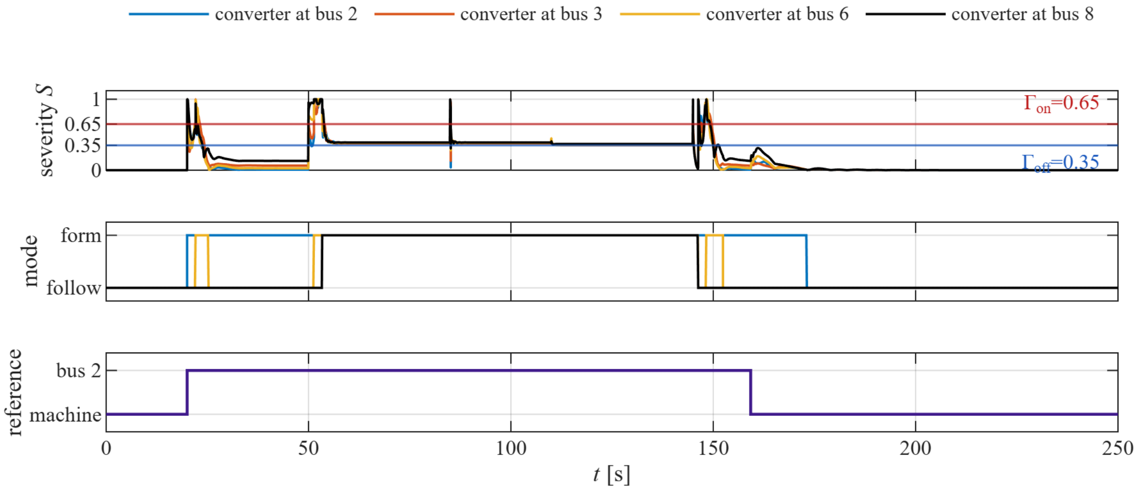
The four-converter set passes this linear screen yet *fails* dynamically: pinned from the trip, sustained power and frequency hunting ends the run fail-closed at $t = 25.488$ s.

The disturbance chronology: six periods



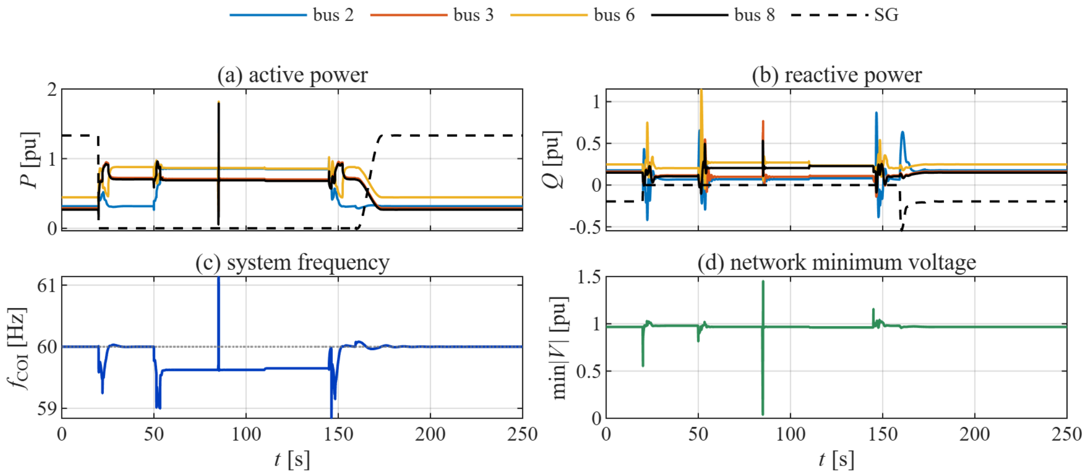
Green: undisturbed. Red: a contingency that removes a source or faults a bus. Yellow: a topology change. Blue: a restorative action. The six scheduled instants are case-defined; the reclose and first release instants are *read from the accepted event log* of the run — they are results, not inputs.

The supervisor’s decision, over the whole run



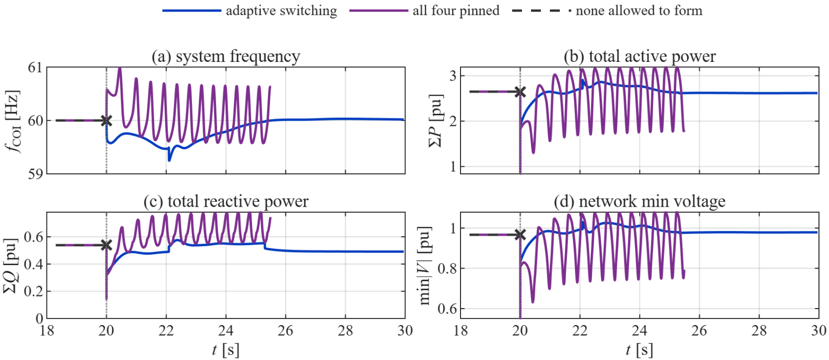
The severity index crosses the commit threshold *only* at the disturbances, and the asymmetric dwell prevents chatter. Peak commitment is 4 converters; the run ends with *none* grid-forming and the machine holding the reference again.

Electrical response on the same time axis



The machine (dashed) drops to zero at the trip and returns at the reclose; the converters take up the difference. Frequency shows both under- and overshoot at the disturbances, then recovers to 60.000001 Hz. Voltage collapses only during the bolted fault and returns to 0.9667–1.0575 p.u..

Switching is necessary, not merely better



None allowed to form — no voltage-forming capability, so the post-trip transaction is *refused* at $t = 20$ s and no trajectory exists. **All four pinned** — rigid forming duty on every unit drives sustained power and frequency hunting. **Staged switching** — rides through all six disturbances.

- f_{COI} settles at nominal 60 Hz without hunting; min $|V|$ recovers after each cleared contingency.
- Settled $|f_{COI} - 60|$ through the fault: 0.3753 Hz staged vs 1.2296 Hz for one pinned forming unit.
- As found: the frozen selections hold the smaller first-event excursion.

Six arms, one chronology

converter policy	horizon [s]	reclose [s]	settled $\lvert f_{COI} - 60 \rvert$ fault window [Hz]	outcome
none allowed to form	20.000	–	–	refused at the trip
one pinned forming	250.000	151.0820	1.2296	completes without adaptation
two pinned forming	250.000	153.9744	0.7126	completes without adaptation
four pinned forming	25.488	–	–	collapse after sustained hunting
switching disabled	250.000	151.0820	1.2296	completes, one unit left forming
staged switching	250.000	159.2519	**0.3753**	**completes; every unit handed back**

Same network, dispatch, event schedule, solver and step control; the arms differ only in the converter policy, and they are numerically identical on the shared window $[0,\ 20.000)$ s — so everything after that instant is attributable to the policy alone. The staged arm terminates at 60.000001 Hz. The fifth arm is the staged run with re-selection switched off: its single commitment coincides with the one-unit pin, which is why the two reproduce each other's peak deviation on every disturbance window, and it ends still holding a forming unit instead of handing it back.

Answers to the three questions

How many?

At least one forming unit is *necessary* for a post-trip trajectory to exist at all. Of 15 candidate sets 7 are admissible, the best margin is an interior pair, and the supervisor stages capacity with the disturbance — up to 4 units here.

Which ones?

Each set is solved at its *own* equilibrium and screened before it can be chosen. With the machine off, the best certified pair is buses $3 + 6$ at $-0.56814\,\mathrm{s}^{-1}$; the ranking decides, not intuition or unit size.

By what route?

Severity-triggered, continuity-gated assignments: quick to promote (0.10 s), deliberately slow to release (1.00 s), each commit tried forward before acceptance. The machine recloses on merit at 159.252 s, the final mode reselection occurs at 173.1271 s, and the hand-back ramp that follows takes 13.8752 s.

One framework, one chronology — measured answers to how many, which ones, and by what route.

Established here — and openly not claimed

Established GATED

- power flow and time-domain results agree with an independent third-party tool on two networks, under tolerances declared in advance — power flow at machine precision
- the composite Jacobian's block structure is derived from the device equations, not assumed as a sparsity pattern
- on this resource mix at least one grid-forming converter is *necessary* for a post-trip trajectory to exist at all
- the admissible grid-forming set is non-monotone in its size, with an interior optimum — so it must be computed, not assumed
- the 250 s chronology closes: the machine recloses on merit, every converter is handed back, and the terminal frequency lands within 10^{-6} Hz of nominal

Not claimed OPEN

- **no operational readiness** — convergence is not grid-code compliance, and the extrema are model excursions, not ratings
- **the dual-mode converter model itself is not cross-validated** against a third-party tool: those gates cover the network and the classical machines only
- **no selector-against-selector** claim: this is switching against *not* switching, on this resource mix
- **no multi-machine** result — that path fails closed and no numbers are assumed in its place
- four of the six severity sub-indices stay diagnostic: they enter no gate and form no aggregate by design
- the DC mode sits at the resolution limit of a phasor method; a firmer link belongs to an electromagnetic-transient study